UNIVERSITI MALAYSIA SARAWAK (UNIMAS)
Faculty of Computer Science and Information Technology

Group Number/Tutorial Slot: Group 13	
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Subject Code: TMF1414	**Subject Name: Introduction To Programming**

Assignment/Project:	**Assignment: UNIMAS PRIMA Outpatient Clinic Billing System**

Name of Lecturer:	**Dr. Irwandi Hipni bin Mohamad Hipiny**

Due Date: 1st December 2025	**Date Submitted: 30th November 2025**

All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor and attached to assignment.

Please note that is your responsibility to retain copies of your assignment.

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Introduction

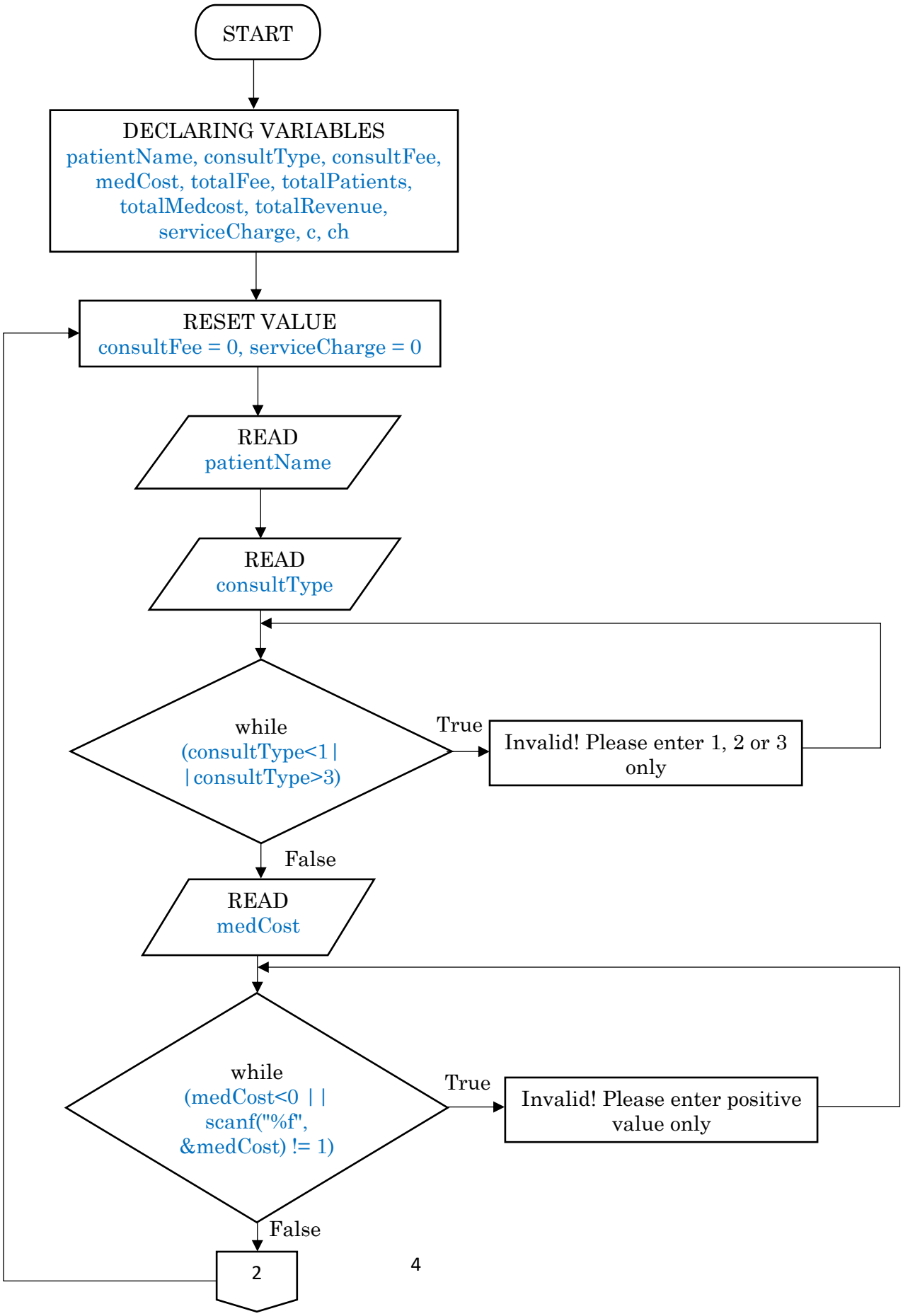
This purpose of this assignment is to create a billing system for UNIMAS PRIMA Outpatient Clinic. The system must be able to calculate the total medical charges for each patient based on their consultation type, medication cost and display the per patient bill details. The system also must allow repetition for multiple patients and show the overall summary of the day which includes number of patients, total medication cost and total collected revenue.

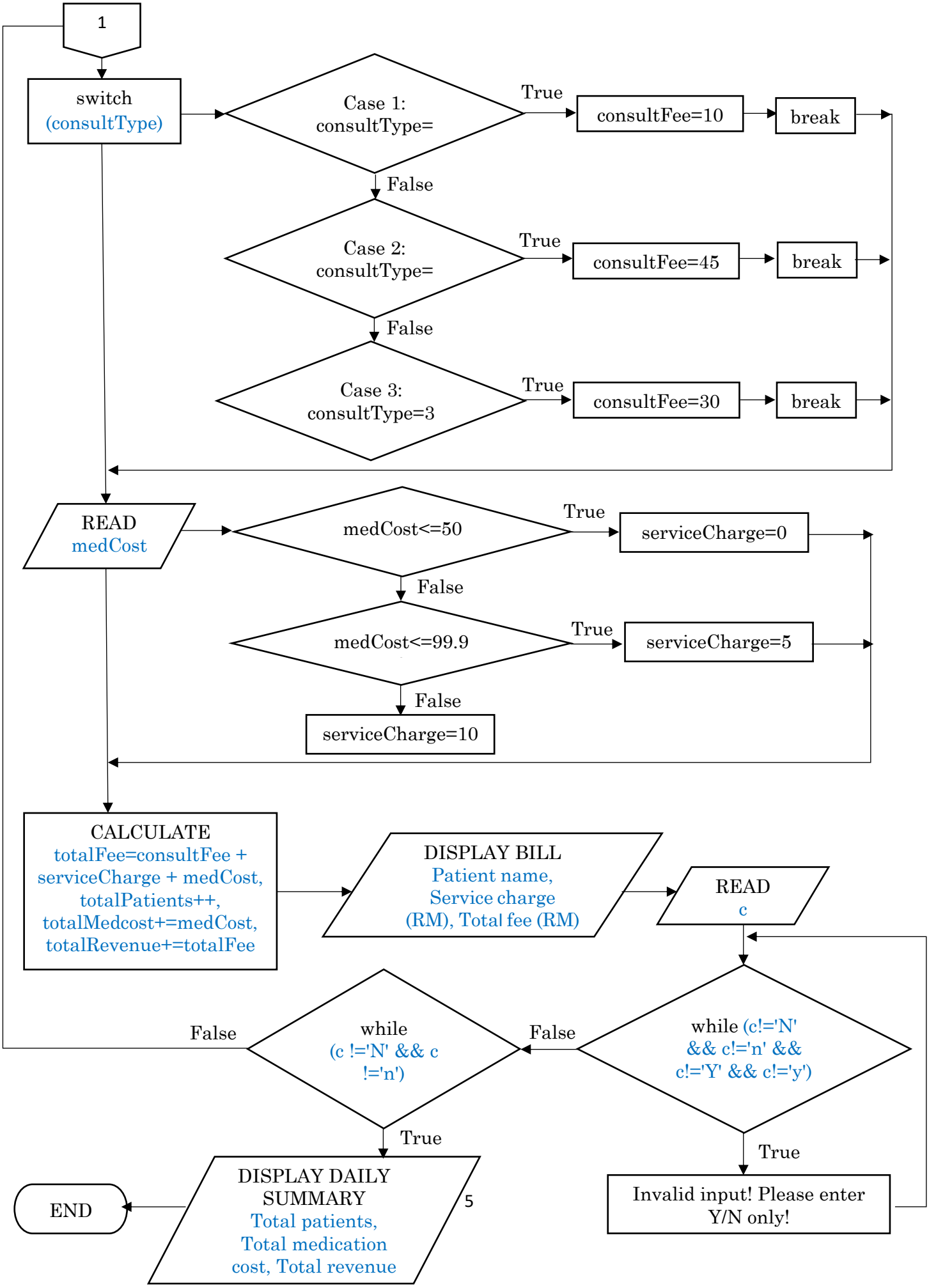
Through this assignment, we need to understand and meet the requirements needed, design the program flow using proper control structures and input validation, and build a working solution using C programming. The program must be organised, readable and produce the correct and clear output. Overall, this assignment strengthens our skills in problem-solving especially in learning on how to turn a real-life situation problem into a working program.

Pseudocode

```
1. START
2. DECLARATION OF VARIABLES
   //For billing
   char patientName[50], c, int consultType, float consultFee = 0.00, medCost,
   totalFee = 0.00, serviceCharge = 0.00
   //For daily summary
   int totalPatients = 0, float totalmedCost = 0.00, totalRevenue = 0.00
   //For input validation
   int ch

3. LOOP STATEMENT BODY (post-test loop: do-while)
   Do
       //Reset values for each next patient
       consultFee = 0, serviceCharge = 0
   3.1 READ INPUT
       //Input patient name
       //Input consultation type (with validation, while: pre-test loop)
       //Input medication cost (with validation, while: pre-test loop)
   3.2 SELECTION STATEMENT (switch)
       switch (consultType)
           case 1: consultFee = 10.00
           case 2: consultFee = 45.00
           case 3: consultFee = 30.00
   3.3 SELECTION STATEMENT (if-else if ladder)
       if (medCost<=50){
            serviceCharge = 0.00;
       } else if (medCost<=99.99){
            serviceCharge = 5.00;
       } else {
            serviceCharge = 10.00;
       }
       //Calculate total fee
       totalFee = consultFee + medCost + serviceCharge
       //Update daily summary
       totalPatients++, totalMedcost += medCost, totalRevenue += totalFee
       //Display bill for each patient
       //Sentinel check ask to continue (with validation, while: pre-test loop)

   } while (c !='N' && c !='n');

4. DISPLAY (daily summary)
5. STOP
```

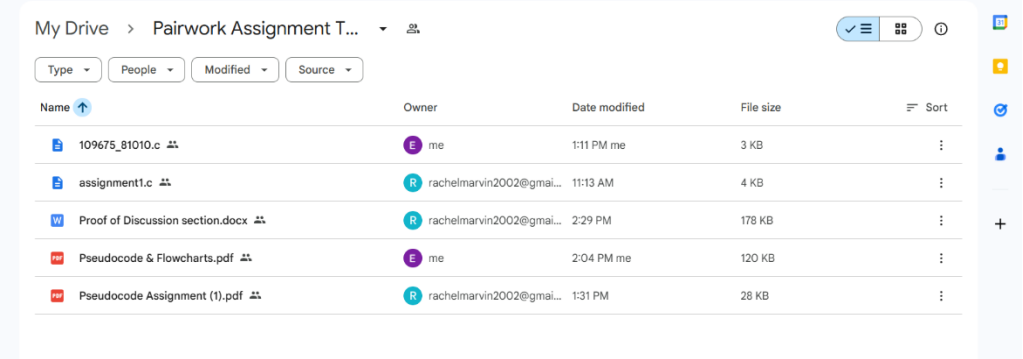
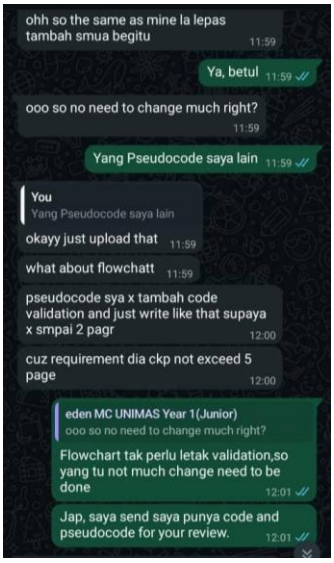
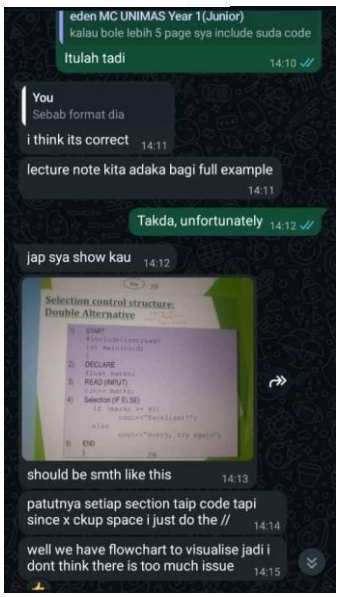
Flowchart


START
DECLARING VARIABLES
patientName, consultType, consultFee, medCost, totalFee, totalPatients, totalMedcost, totalRevenue, serviceCharge, c, ch
RESET VALUE
consultFee = 0, serviceCharge = 0
READ
patientName
READ
consultType
while
(consultType<1 | | consultType>3)
True
Invalid! Please enter 1, 2 or 3 only
False
READ
medCost
while
(medCost<0 | | scanf("%f", &medCost) != 1)
True
Invalid! Please enter positive value only
False
2


1
switch (consultType)
Case 1: consultType=
True
consultFee=10
break
False
Case 2: consultType=
True
consultFee=45
break
False
Case 3: consultType=3
True
consultFee=30
break
READ medCost
medCost<=50
True
serviceCharge=0
False
medCost<=99.9
True
serviceCharge=5
False
serviceCharge=10
CALCULATE totalFee=consultFee + serviceCharge + medCost, totalPatients++, totalMedcost+=medCost, totalRevenue+=totalFee
DISPLAY BILL Patient name, Service charge (RM), Total fee (RM)
READ c
while (c!='N' && c!='n' && c!='Y' && c!='y')
False
True
Invalid input! Please enter Y/N only!
while (c !='N' && c !='n')
False
True
DISPLAY DAILY SUMMARY Total patients, Total medication cost, Total revenue
END

Sample Run of Program

- The code run validation test on consultation type, medication cost and the sentinel (Y/N) to ensure the user only enter the correct input format.
- The code displays each bill per patient after the user enter all the details.
- Daily summary is displayed at the end after the looping stop when user enter sentinel 'N' or 'n'.

```
---------Welcome To UNIMAS PRIMA Patient Billing System---------
Enter Patient Name: Ali
Select Consultation Type (1=General, 2=Specialist, 3=Emergency): 0
Invalid format of consultation type. Please enter enter either 1, 2 or 3 only: a
Invalid format of consultation type. Please enter enter either 1, 2 or 3 only: 4
Invalid format of consultation type. Please enter enter either 1, 2 or 3 only: 3
Enter medication cost (RM): -1
Invalid format of medication cost. Enter positive value only: a
Invalid format of medication cost. Enter positive value only: 150

---------Patient Bill---------
Patient name: Ali
Service charge (RM): 10.00
Total Fee (RM): RM190.00
------------------------------

Continue for another patient? (Y/N): a
Invalid input! Please enter Y/N only!: 3
Invalid input! Please enter Y/N only!: y

---------Welcome To UNIMAS PRIMA Patient Billing System---------
Enter Patient Name: Abu
Select Consultation Type (1=General, 2=Specialist, 3=Emergency): 3
Enter medication cost (RM): 100

---------Patient Bill---------
Patient name: Abu
Service charge (RM): 10.00
Total Fee (RM): RM140.00
------------------------------

Continue for another patient? (Y/N): n

---------Daily Summary------------
Total patients: 2
Total Medication Cost: 250.00
Total Revenue: 330.00
-----------------------------------
Process returned 0 (0x0)   execution time : 66.371 s
Press any key to continue.
```

Proof of Discussion